



Unit II: Discrete Distributions

Uniform, Bernoulli, Binomial, Geometric,
Negative Binomial, Poisson, Hypergeometric

STA201 – Theory of Probability and Statistics

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1 Motivation: Why Study Probability Distributions?

In the previous section, we introduced random variables and their probability mass functions (PMFs). For any discrete random variable X , the PMF $p(x) = P(X = x)$ describes completely how probability is distributed over the possible values of X .

In engineering practice, however, we frequently encounter situations that share the same underlying probabilistic structure. For example:

- Inspecting 50 items from a production line and counting defectives.
- Counting the number of phone calls arriving at a server in one minute.
- Testing components until the first failure occurs.
- Selecting 5 cards from a deck and counting aces.

These seemingly different experiments can be modelled by a small number of **standard probability distributions**. Knowing the distribution allows us to:

1. Compute probabilities quickly using standard formulas.
2. Obtain mean and variance without lengthy summations.
3. Make predictions and decisions in engineering design and quality control.

In this unit, we study the most important **discrete distributions** in the following order:

1. Discrete Uniform
2. Bernoulli
3. Binomial
4. Geometric
5. Negative Binomial
6. Poisson
7. Hypergeometric

This ordering is deliberate: it follows a natural progression from the simplest distribution to more complex ones, with each new distribution building on the previous. At the end of the unit, we summarize the connections between them.

2 Discrete Uniform Distribution

2.1 Motivation

The simplest discrete distribution arises when a random variable takes a finite number of values, each with equal probability. Examples include:

- Rolling a fair die: $X \in \{1, 2, 3, 4, 5, 6\}$, each with probability $1/6$.
- Randomly selecting one item from a batch of n items, all equally likely.
- Choosing a random number from $\{1, 2, \dots, N\}$.

2.2 Definition

A random variable X has a **discrete uniform distribution** on the set $\{a, a + 1, \dots, b\}$ if

$$P(X = x) = \frac{1}{b - a + 1}, \quad x = a, a + 1, \dots, b.$$

Here $n = b - a + 1$ is the number of possible values.

2.3 Mean and Variance

$$E[X] = \frac{a+b}{2},$$

$$\text{Var}(X) = \frac{(b-a+1)^2 - 1}{12} = \frac{n^2 - 1}{12}.$$

Solved Example Die roll

A fair six-sided die is rolled. Let X be the face value. Here $a = 1, b = 6, n = 6$.

$$E[X] = \frac{1+6}{2} = 3.5, \quad \text{Var}(X) = \frac{6^2 - 1}{12} = \frac{35}{12} \approx 2.917.$$

This matches our earlier calculation from first principles.

Solved Example Random selection of components

A batch contains 10 components labelled 1 to 10. One component is selected at random. Let X be the label number. Find $P(X \geq 7)$, $E[X]$, and $\text{Var}(X)$.

Solution: Here $a = 1, b = 10, n = 10$.

$$P(X \geq 7) = \frac{10 - 7 + 1}{10} = \frac{4}{10} = 0.4.$$

$$E[X] = \frac{1+10}{2} = 5.5, \quad \text{Var}(X) = \frac{10^2 - 1}{12} = \frac{99}{12} = 8.25.$$

Solved Example Uniform on a different range

A random variable X is uniformly distributed on $\{3, 4, 5, 6, 7, 8\}$. Find $P(X \leq 5)$, $E[X]$, and $\text{Var}(X)$.

Solution: Here $a = 3, b = 8, n = 6$.

$$P(X \leq 5) = \frac{3}{6} = 0.5.$$

$$E[X] = \frac{3+8}{2} = 5.5, \quad \text{Var}(X) = \frac{6^2 - 1}{12} = \frac{35}{12} \approx 2.917.$$

3 Bernoulli Distribution

3.1 Motivation

Many engineering experiments have only two possible outcomes: pass/fail, success/failure, defective/non-defective, yes/no. The Bernoulli distribution models a single trial of this kind. It is the fundamental building block of the binomial, geometric, and negative binomial distributions.

3.2 Definition

A random variable X has a **Bernoulli distribution** with parameter p ($0 \leq p \leq 1$) if

$$P(X = 1) = p, \quad P(X = 0) = 1 - p = q.$$

We write $X \sim \text{Bernoulli}(p)$. Equivalently,

$$P(X = x) = p^x q^{1-x}, \quad x = 0, 1.$$

3.3 Mean and Variance

$$\begin{aligned} E[X] &= p, \\ \text{Var}(X) &= p(1 - p) = pq. \end{aligned}$$

Solved Example Single component test

A component is tested and is defective with probability 0.05. Let $X = 1$ if defective, $X = 0$ if non-defective. Find $E[X]$ and $\text{Var}(X)$.

Solution:

$$E[X] = 0.05, \quad \text{Var}(X) = 0.05 \times 0.95 = 0.0475.$$

Solved Example Relay functioning

The probability that a relay will function properly in a control system is 0.98. Let $X = 1$ if it functions and $X = 0$ otherwise. Find the mean and variance of X .

Solution:

$$E[X] = 0.98, \quad \text{Var}(X) = 0.98 \times 0.02 = 0.0196.$$

Solved Example Bit transmission

A single bit is transmitted through a noisy channel. The probability of correct reception is 0.999. Let $X = 1$ if the bit is received correctly. Find $E[X]$ and $\text{Var}(X)$.

Solution:

$$E[X] = 0.999, \quad \text{Var}(X) = 0.999 \times 0.001 = 0.000999.$$

3.4 Connection to Later Distributions

The Bernoulli distribution is the simplest case of the binomial distribution ($n = 1$). If we repeat independent Bernoulli trials, we obtain the Bernoulli process, which gives rise to the binomial, geometric, and negative binomial distributions depending on what we count.

4 Binomial Distribution

4.1 Motivation

The binomial distribution arises when we perform n independent Bernoulli trials, each with the same success probability p , and count the total number of successes. Engineering applications:

- Number of defective items in a batch of n .
- Number of bits received in error in a packet of n bits.
- Number of components that survive a stress test.
- Number of patients responding positively to a treatment.

4.2 Definition

A random variable X has a **binomial distribution** with parameters n (number of trials) and p (success probability) if

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n.$$

We write $X \sim B(n, p)$.

4.3 Derivation from Bernoulli Trials

Consider a specific sequence of n trials with exactly k successes and $n - k$ failures. Because the trials are independent, the probability of this particular sequence is

$$p^k (1 - p)^{n-k}.$$

The number of such sequences is the number of ways to choose which k of the n trials are successes:

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

Since the sequences are mutually exclusive, we add their probabilities:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

4.4 Mean and Variance

$$\begin{aligned} E[X] &= np, \\ \text{Var}(X) &= np(1 - p) = npq. \end{aligned}$$

Solved Example Defective items in a batch

A manufacturing process produces items with a defective rate of 2%. A batch of 20 items is selected. Let X be the number of defectives. Find $P(X = 0)$, $P(X \leq 1)$, and $E[X]$.

Solution: $X \sim B(20, 0.02)$.

$$P(X = 0) = \binom{20}{0} (0.02)^0 (0.98)^{20} = (0.98)^{20} \approx 0.6676.$$

$$P(X = 1) = \binom{20}{1} (0.02)(0.98)^{19} = 20 \times 0.02 \times (0.98)^{19} \approx 0.2725.$$

$$P(X \leq 1) = P(X = 0) + P(X = 1) \approx 0.6676 + 0.2725 = 0.9401.$$

$$E[X] = np = 20 \times 0.02 = 0.4.$$

Solved Example Communication channel

A binary communication channel transmits bits with a bit error probability $p = 0.001$. A packet contains 1000 bits. Let X be the number of errors. Find $P(X = 0)$, $P(X = 1)$, and $P(X \geq 2)$.

Solution: $X \sim B(1000, 0.001)$.

$$P(X = 0) = (0.999)^{1000} \approx 0.3677.$$

$$P(X = 1) = \binom{1000}{1} (0.001)(0.999)^{999} \approx 0.3681.$$

$$P(X \geq 2) = 1 - 0.3677 - 0.3681 = 0.2642.$$

Solved Example Quality control

A production line has a 95% yield (non-defective). A random sample of 10 items is drawn. Find the probability that exactly 8 are non-defective.

Solution: Here $p = 0.95$, $n = 10$, $k = 8$.

$$P(X = 8) = \binom{10}{8} (0.95)^8 (0.05)^2 = 45 \times 0.6634 \times 0.0025 \approx 0.0746.$$

Solved Example Engine reliability

An aircraft has 4 engines. The probability that an engine fails on a given flight is 0.01, and engines fail independently. The aircraft can fly safely if at least 2 engines are working. Find the probability that a flight is safe.

Solution: Let X be the number of engines that fail. Then $X \sim B(4, 0.01)$. The flight is safe if $X \leq 2$.

$$P(X = 0) = (0.99)^4 \approx 0.9606.$$

$$P(X = 1) = \binom{4}{1} (0.01)(0.99)^3 \approx 0.0388.$$

$$P(X = 2) = \binom{4}{2} (0.01)^2 (0.99)^2 \approx 0.0006.$$

$$P(\text{safe}) = P(X \leq 2) \approx 0.9606 + 0.0388 + 0.0006 = 1.0000 \text{ (approximately).}$$

Solved Example Multiple choice test

A student guesses randomly on a 10-question multiple choice test where each question has 4 choices. Find the probability of getting exactly 5 correct answers.

Solution: Here $p = 1/4 = 0.25$, $n = 10$, $k = 5$.

$$P(X = 5) = \binom{10}{5} (0.25)^5 (0.75)^5 = 252 \times 0.0009766 \times 0.2373 \approx 0.0584.$$

Solved Example Probability of at least one defective

A factory produces items with a 3% defective rate. If 30 items are inspected, find the probability that at least one is defective.

Solution: $X \sim B(30, 0.03)$.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.97)^{30} \approx 1 - 0.4010 = 0.5990.$$

4.5 Properties of the Binomial Distribution

- **Symmetry:** If $p = 0.5$, the PMF is symmetric about $n/2$.
- **Additivity:** If $X_1 \sim B(n_1, p)$ and $X_2 \sim B(n_2, p)$ are independent, then

$$X_1 + X_2 \sim B(n_1 + n_2, p).$$

- **Recurrence relation:**

$$\frac{P(X = k + 1)}{P(X = k)} = \frac{n - k}{k + 1} \cdot \frac{p}{1 - p}.$$

- **Mode:** The most probable value is $\lfloor (n + 1)p \rfloor$.

5 Geometric Distribution

5.1 Motivation

The geometric distribution is also built on a Bernoulli process. Instead of fixing the number of trials n in advance and counting successes, we ask a different question: **how many trials are needed to get the first success?** Engineering examples:

- Number of components tested until the first defective one is found.
- Number of transmission attempts until a packet is successfully received.
- Number of inspections until the first fault is detected.
- Number of calls until the first successful connection.

5.2 Definition

A random variable X has a **geometric distribution** with success probability p if

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

Here X counts the trial on which the first success occurs.

5.3 Mean and Variance

$$E[X] = \frac{1}{p},$$

$$\text{Var}(X) = \frac{1-p}{p^2} = \frac{q}{p^2}.$$

5.4 Derivation of the Mean

$$E[X] = \sum_{k=1}^{\infty} kq^{k-1}p = p \sum_{k=1}^{\infty} kq^{k-1}.$$

Using $\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2} = \frac{1}{p^2}$,

$$E[X] = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

5.5 Memorylessness Property

The geometric distribution is the only discrete distribution with the **memoryless property**:

$$P(X > m + n \mid X > m) = P(X > n).$$

This means that if you have already failed m times, the probability of needing n more trials to succeed is the same as if you were starting fresh.

Solved Example First defective item

A production line produces items with a defective rate of 10%. Items are inspected until the first defective is found. Find the probability that the first defective is found on the 3rd inspection.

Solution: $X \sim \text{Geometric}(0.1)$.

$$P(X = 3) = (0.9)^2(0.1) = 0.81 \times 0.1 = 0.081.$$

The expected number of inspections to find the first defective is $1/0.1 = 10$.

Solved Example Packet retransmission

A wireless link successfully delivers a packet with probability 0.8 on each attempt. Attempts are independent. Let X be the number of attempts needed for the first successful delivery. Find $P(X \leq 3)$.

Solution: $X \sim \text{Geometric}(0.8)$.

$$\begin{aligned} P(X \leq 3) &= P(X = 1) + P(X = 2) + P(X = 3). \\ &= 0.8 + (0.2)(0.8) + (0.2)^2(0.8) = 0.8 + 0.16 + 0.032 = 0.992. \end{aligned}$$

Solved Example Coin tossing

Find the probability that a person flipping a fair coin gets the first head on the fourth trial.

Solution: $p = 0.5$. $X \sim \text{Geometric}(0.5)$.

$$P(X = 4) = (0.5)^3(0.5) = (0.5)^4 = 0.0625.$$

Solved Example Expected waiting time

A quality engineer inspects items with a 5% defective rate. Find the expected number of inspections to find the first defective and the variance.

Solution:

$$E[X] = \frac{1}{0.05} = 20, \quad \text{Var}(X) = \frac{0.95}{(0.05)^2} = \frac{0.95}{0.0025} = 380.$$

$$\sigma_X = \sqrt{380} \approx 19.49.$$

Solved Example Memorylessness

Suppose $X \sim \text{Geometric}(0.2)$. Find $P(X > 5 \mid X > 3)$.

Solution: By memorylessness,

$$P(X > 5 \mid X > 3) = P(X > 2) = (0.8)^2 = 0.64.$$

5.6 Connection to the Binomial Distribution

Both distributions arise from the same Bernoulli process:

- **Binomial:** Fix n , count successes.
- **Geometric:** Fix 1 success, count trials.

6 Negative Binomial Distribution

6.1 Motivation

The negative binomial distribution generalises the geometric distribution: instead of waiting for the *first* success, we wait for the r -th success. Engineering examples:

- Number of items inspected until the 3rd defective is found.
- Number of transmissions needed to deliver 5 packets successfully.
- Number of tests until 2 faults are detected.

6.2 Definition

A random variable X has a **negative binomial distribution** with parameters r (number of successes) and p (success probability) if

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \dots$$

Here X counts the trial on which the r -th success occurs. For $r = 1$, it reduces to the geometric distribution.

6.3 Mean and Variance

$$E[X] = \frac{r}{p},$$

$$\text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{rq}{p^2}.$$

6.4 Derivation of the Mean

Write $X = X_1 + X_2 + \dots + X_r$, where each X_i is the number of trials needed for the i -th success after the $(i-1)$ -th success. Each X_i is geometric with parameter p and mean $1/p$. By linearity of expectation,

$$E[X] = r \cdot \frac{1}{p} = \frac{r}{p}.$$

Similarly, since the X_i are independent, $\text{Var}(X) = r \cdot \frac{q}{p^2} = \frac{rq}{p^2}$.

Solved Example Third defective item

A process has a defective rate of 5%. Items are inspected until 3 defectives are found. Find the probability that the 3rd defective occurs on the 20th inspection.

Solution: Here $r = 3, p = 0.05, k = 20$.

$$P(X = 20) = \binom{19}{2} (0.05)^3 (0.95)^{17}.$$

$$\binom{19}{2} = 171, \quad (0.05)^3 = 0.000125, \quad (0.95)^{17} \approx 0.4181.$$

$$P(X = 20) = 171 \times 0.000125 \times 0.4181 \approx 0.00894.$$

The expected number of inspections to find 3 defectives is $r/p = 3/0.05 = 60$.

Solved Example Third head on seventh toss

Find the probability that a person flipping a fair coin gets the third head on the seventh trial.

Solution: Here $r = 3, p = 0.5, k = 7$.

$$P(X = 7) = \binom{6}{2} (0.5)^3 (0.5)^4 = 15 \times 0.125 \times 0.0625 = 15 \times 0.0078125 = 0.1172.$$

Solved Example Packet delivery

A wireless link delivers packets successfully with probability 0.9. Find the probability that the 4th successful delivery occurs on the 6th transmission.

Solution: Here $r = 4, p = 0.9, k = 6$.

$$P(X = 6) = \binom{5}{3} (0.9)^4 (0.1)^2 = 10 \times 0.6561 \times 0.01 = 0.06561.$$

The expected number of transmissions is $r/p = 4/0.9 \approx 4.44$.

Solved Example Expected inspections

A quality engineer inspects items with a 10% defective rate until 5 defectives are found. Find the expected number of inspections and the variance.

Solution:

$$E[X] = \frac{5}{0.1} = 50, \quad \text{Var}(X) = \frac{5 \times 0.9}{(0.1)^2} = \frac{4.5}{0.01} = 450.$$

6.5 Connection to Geometric and Binomial

- **Geometric:** Special case of negative binomial with $r = 1$.
- **Binomial:** Both arise from Bernoulli process; binomial fixes n and counts successes, negative binomial fixes r successes and counts trials.

7 Poisson Distribution**7.1 Motivation**

The Poisson distribution models the number of events occurring in a fixed interval of time, length, area, or volume, when events occur randomly and independently at a constant average rate. Engineering examples:

- Number of arrivals at a server per minute.
- Number of defects per square metre of fabric.
- Number of particles emitted by a radioactive source per second.
- Number of lightning strikes on a power line per year.

7.2 Definition

A random variable X has a **Poisson distribution** with parameter $\lambda > 0$ if

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

We write $X \sim \text{Poisson}(\lambda)$. Here λ is the average rate of occurrence.

7.3 Mean and Variance

$$\begin{aligned} E[X] &= \lambda, \\ \text{Var}(X) &= \lambda. \end{aligned}$$

Note the elegant property: for the Poisson distribution, **mean equals variance**.

7.4 The Poisson Process

The Poisson distribution is intimately connected with the **Poisson process**, a model for events occurring randomly in continuous time. If events occur at rate λ per unit time, then the number of events X in a time interval of length t has a Poisson distribution with parameter λt :

$$P(X = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}.$$

Solved Example Server arrivals

Calls arrive at a server at an average rate of 4 per minute. Find the probability that exactly 2 calls arrive in a given minute.

Solution: $X \sim \text{Poisson}(4)$.

$$P(X = 2) = \frac{e^{-4} 4^2}{2!} = \frac{e^{-4} \times 16}{2} = 8e^{-4} \approx 0.1465.$$

Solved Example Defects per unit area

A textile mill finds an average of 2 defects per square metre of fabric. Find the probability that a randomly chosen square metre has no defect.

Solution: $X \sim \text{Poisson}(2)$.

$$P(X = 0) = e^{-2} \approx 0.1353.$$

Solved Example Poisson approximation to binomial

A communication link transmits $n = 10000$ bits with bit error probability $p = 10^{-4}$. Use the Poisson approximation to find $P(X = 0)$, $P(X = 1)$, and $P(X \geq 2)$.

Solution: Here $\lambda = np = 10000 \times 10^{-4} = 1$.

$$P(X = 0) = e^{-1} \approx 0.3679, \quad P(X = 1) = e^{-1} \approx 0.3679.$$

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) \approx 0.2642.$$

These values are very close to the exact binomial probabilities. The Poisson approximation is excellent when n is large and p is small.

Solved Example Radioactive decay

The average number of radioactive particles passing through a counter in 1 millisecond is 4. Find the probability that 6 particles enter the counter in a given millisecond.

Solution: $X \sim \text{Poisson}(4)$.

$$P(X = 6) = \frac{e^{-4}4^6}{6!} = \frac{e^{-4} \times 4096}{720} \approx 0.1042.$$

Solved Example Oil tankers

Ten is the average number of oil tankers arriving each day at a certain port. The facilities can handle at most 15 tankers per day. What is the probability that on a given day tankers will have to be turned away?

Solution: $X \sim \text{Poisson}(10)$. Tankers are turned away if $X > 15$.

$$P(X > 15) = 1 - P(X \leq 15) = 1 - \sum_{k=0}^{15} \frac{e^{-10}10^k}{k!} \approx 1 - 0.9513 = 0.0487.$$

So there is about a 4.87% chance that tankers will be turned away.

Solved Example Poisson process with varying interval

If calls arrive at a server according to a Poisson process with rate 4 per minute, find the probability that exactly 3 calls arrive in a 2-minute interval.

Solution: Here $\lambda = 4$, $t = 2$, so $\lambda t = 8$.

$$P(X = 3) = \frac{e^{-8}8^3}{3!} = \frac{e^{-8} \times 512}{6} \approx 0.0286.$$

Solved Example Flaws in copper wire

Flaws in a copper wire occur according to a Poisson process with an average of 0.5 flaws per metre. Find the probability that a 3-metre length contains at least one flaw.

Solution: Here $\lambda = 0.5$, $t = 3$, so $\lambda t = 1.5$.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-1.5} \approx 1 - 0.2231 = 0.7769.$$

7.5 Connection to the Binomial Distribution

The Poisson distribution is the limiting case of the binomial distribution when:

$$n \rightarrow \infty, \quad p \rightarrow 0, \quad np = \lambda \text{ (constant)}.$$

This is why the Poisson distribution is often called the **law of rare events**. It is also the continuous-time analogue of the binomial distribution: instead of counting successes in a fixed number of discrete trials, we count events in a fixed continuous interval.

8 Hypergeometric Distribution

8.1 Motivation

The hypergeometric distribution models sampling **without replacement** from a finite population consisting of two types of items. It differs from the binomial distribution, which assumes independent trials (sampling with replacement). Engineering examples:

- Number of defective items in a sample drawn without replacement from a finite batch.
- Number of acceptable components in a lot inspected without replacement.
- Number of aces in a poker hand.

8.2 Definition

Suppose a population of size N contains K items of one type (called “successes”) and $N - K$ of the other. A sample of size n is drawn without replacement. Let X be the number of successes in the sample. Then

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad k = \max(0, n - (N - K)), \dots, \min(n, K).$$

We write $X \sim \text{Hypergeometric}(N, K, n)$.

8.3 Mean and Variance

$$E[X] = n \frac{K}{N},$$

$$\text{Var}(X) = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1}.$$

The factor $\frac{N-n}{N-1}$ is called the **finite population correction factor**. When n is small compared to N , this factor is close to 1, and the hypergeometric distribution is well approximated by the binomial with $p = K/N$.

Solved Example Defective items without replacement

A batch contains 20 items, of which 5 are defective. Three items are drawn at random without replacement. Let X be the number of defectives drawn. Find the PMF of X and $P(X = 2)$.

Solution: Here $N = 20, K = 5, n = 3$.

$$P(X = k) = \frac{\binom{5}{k} \binom{15}{3-k}}{\binom{20}{3}}, \quad k = 0, 1, 2, 3.$$

For $k = 2$:

$$P(X = 2) = \frac{\binom{5}{2} \binom{15}{1}}{\binom{20}{3}} = \frac{10 \times 15}{1140} = \frac{150}{1140} \approx 0.1316.$$

The mean is $E[X] = nK/N = 3 \times 5/20 = 0.75$.

Solved Example Poker hand

In a 5-card poker hand from a standard 52-card deck, find the probability of holding exactly 2 aces.

Solution: Here $N = 52, K = 4$ (aces), $n = 5, k = 2$.

$$P(X = 2) = \frac{\binom{4}{2} \binom{48}{3}}{\binom{52}{5}} = \frac{6 \times 17296}{2598960} \approx 0.0399.$$

Solved Example Hypergeometric mean and variance

A quality engineer draws 10 items without replacement from a batch of 100, of which 8 are defective. Find the expected number of defectives and the variance.

Solution: $N = 100, K = 8, n = 10$.

$$E[X] = n \frac{K}{N} = 10 \times \frac{8}{100} = 0.8.$$

$$\text{Var}(X) = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1} = 10 \times 0.08 \times 0.92 \times \frac{90}{99} \approx 0.6691.$$

Compare with the binomial variance $npq = 10(0.08)(0.92) = 0.736$, which is slightly larger because the finite population correction factor is less than 1.

Solved Example Missiles

From a lot of 10 missiles, 4 are selected at random and fired. If the lot contains 3 defective missiles that will not fire, what is the probability that at most 2 will not fire?

Solution: Here $N = 10$, $K = 3$ (defective), $n = 4$. We want $P(X \leq 2)$.

$$P(X = 0) = \frac{\binom{3}{0} \binom{7}{4}}{\binom{10}{4}} = \frac{1 \times 35}{210} = \frac{35}{210} \approx 0.1667.$$

$$P(X = 1) = \frac{\binom{3}{1} \binom{7}{3}}{\binom{10}{4}} = \frac{3 \times 35}{210} = \frac{105}{210} = 0.5.$$

$$P(X = 2) = \frac{\binom{3}{2} \binom{7}{2}}{\binom{10}{4}} = \frac{3 \times 21}{210} = \frac{63}{210} = 0.3.$$

$$P(X \leq 2) = 0.1667 + 0.5 + 0.3 = 0.9667.$$

8.4 Connection to the Binomial Distribution

When the population size N is very large compared to the sample size n , the hypergeometric distribution is well approximated by the binomial distribution with $p = K/N$. This is because drawing without replacement from a large population has nearly the same effect as drawing with replacement.

9 Unified View: Connections Between Discrete Distributions

The distributions we have studied are not isolated; they are connected in a beautiful family tree rooted in the Bernoulli process.

The Bernoulli Process as the Foundation

A Bernoulli process is a sequence of independent trials, each with two outcomes (success with probability p , failure with probability $q = 1 - p$). From this single process, we obtain:

- **Bernoulli:** Outcome of a single trial.
- **Binomial:** Number of successes in a fixed number of trials n .
- **Geometric:** Number of trials until the first success.
- **Negative Binomial:** Number of trials until the r -th success.

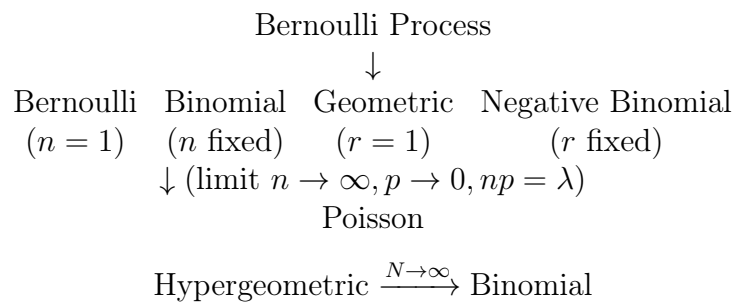
The Poisson Limit

If we take the binomial distribution and let $n \rightarrow \infty$, $p \rightarrow 0$, with $np = \lambda$ constant, we obtain the **Poisson distribution**. This is the continuous-time analogue of the Bernoulli process.

The Hypergeometric Connection

The hypergeometric distribution arises when sampling without replacement from a finite population. As the population size becomes large, it converges to the binomial distribution.

Summary Diagram



10 Summary Table of Discrete Distributions

Table 1: Summary of discrete distributions

Distribution	PMF	Mean	Variance
Uniform(a, b)	$\frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{n^2-1}{12}$
Bernoulli(p)	$p^k q^{1-k}$	p	pq
Binomial(n, p)	$\binom{n}{k} p^k q^{n-k}$	np	npq
Geometric(p)	$q^{k-1} p$	$\frac{1}{p}$	$\frac{q}{p^2}$
Neg. Binomial(r, p)	$\binom{k-1}{r-1} p^r q^{k-r}$	$\frac{r}{p}$	$\frac{rq}{p^2}$
Poisson(λ)	$\frac{e^{-\lambda} \lambda^k}{k!}$	λ	λ
Hypergeometric	$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$	$n \frac{K}{N}$	$n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$

11 When to Use Which Distribution

Table 2: Choosing the right discrete distribution

Situation	Distribution	Key Question
Finite equally likely outcomes	Uniform	Each value has equal probability?
Single success/failure trial	Bernoulli	One trial, two outcomes?
Fixed number of independent trials	Binomial	How many successes in n trials?
Waiting for first success	Geometric	How many trials until first success?
Waiting for r -th success	Negative Binomial	How many trials until r successes?
Events in a fixed interval	Poisson	How many events per unit time/space?
Sampling without replacement from finite population	Hypergeometric	How many successes in sample without replacement?

12 Practice Problems

1. A fair die is rolled 10 times. Find the probability of obtaining exactly three 6's.
2. A production line has a 3% defective rate. Find the probability that in a sample of 25 items, exactly 2 are defective. Also find the expected number of defectives.
3. A component has a 90% probability of functioning correctly. Components are tested until the first failure. Find the probability that the first failure occurs on the 5th test and the expected number of tests.
4. Arrivals at a service counter occur at an average rate of 5 per hour. Find the probability that exactly 3 arrivals occur in a given hour, and the probability that no arrival occurs.
5. A lot of 50 items contains 10 defective items. A sample of 8 is drawn without replacement. Find the probability that exactly 2 are defective.
6. Packets arrive at a router at an average rate of 8 per second. Find the probability that more than 10 packets arrive in a given second.
7. In a batch of 15 batteries, 4 are faulty. Three batteries are selected at random without replacement. Find the probability that at most one is faulty.
8. A coin is tossed until 4 heads appear. If $P(H) = 0.6$, find the probability that the 4th head occurs on the 8th toss.
9. A multiple choice test has 20 questions, each with 5 possible answers. A student guesses randomly. Find the probability that the student gets more than 5 correct answers.

10. The number of typing errors on a page follows a Poisson distribution with a mean of 2 errors per page. Find the probability that a randomly selected page has exactly 3 errors.
11. A bag contains 20 red and 30 blue marbles. Ten marbles are drawn without replacement. Find the probability that exactly 4 are red.
12. An urn contains 10 white and 8 black balls. Balls are drawn one at a time with replacement until a white ball is drawn. Find the probability that the first white ball is drawn on the 4th draw.
13. A random variable X is uniformly distributed on $\{2, 3, 4, 5, 6\}$. Find $E[X]$ and $\text{Var}(X)$.
14. If $X \sim B(20, 0.4)$, find $E[X]$, $\text{Var}(X)$, and the mode of X .
15. A communication channel has a bit error probability of 10^{-3} . If 2000 bits are transmitted, find the probability that more than 3 bits are in error. Use the Poisson approximation.
16. A batch of 50 components is tested. Each component fails with probability 0.01. Find the probability that exactly 2 fail.
17. A basketball player makes a free throw with probability 0.7. If she takes 10 free throws, find the probability that she makes at least 8.
18. A quality inspector samples 100 items from a large lot with a 5% defective rate. Find the probability that exactly 5 are defective.
19. Suppose $X \sim \text{Geometric}(0.3)$. Find $P(X > 4)$.
20. A production process has a 2% defective rate. Items are inspected until 5 defectives are found. Find the expected number of inspections and the variance.